Abridged Syllabus: Introduction to Discrete Structures

Contents

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1 Details

Course & Section:	*Introduction to Discrete Structures,* CISC 2210, MW3
Days & Time:	Mondays, Wednesdays (MoWe), 03:40 PM – 04:55 PM
Location:	Ingersoll Hall Extension, Room 4141 (IA-4141 in short)
Instructor:	Miriam Briskman
E-mail:	miriam.briskman@brooklyn.cuny.edu
Response Time:	Within 24 – 48 hours, between 12 PM to 9:30 PM
Office Hours:	Wednesdays, 07:30 PM – 09:30 PM, online through Zoom. You can find the link on Brightspace, in Content → Office Hours. Alternatively, please email the instructor to schedule an appointment.
Course Materials:	[Free] *Discrete Math*, by Mohamed Jamaloodee et al. Link: `https://ggc-discrete-math.github.io/`. [Free] *A Cool Brisk Walk Through Discrete Mathematics*, by Davis Stephen. Link: `http://stephendavies.org/brisk.pdf`. [Free] *Introduction to Discrete Mathematics: An OER for MA-471*, by Mathieu Sassolas. Link: `https://academicworks.cuny.edu/cgi/viewcontent.cgi?article=1183&context=qb_oers`. [Free] *Discrete Mathematics: An Open Introduction*, by Oscar Levin. Link: `https://discrete.openmathbooks.org/dmoi3/`. [Free] *Notes on Discrete Mathematics*, by James Aspnes. Link: `https://www.cs.yale.edu/homes/aspnes/classes/202/notes.pdf`. [Free] *Applied Discrete Structures*, by Al Doerr and Ken Levasseur. Link: `https://discretemath.org/ads-latex/ads.pdf`. [Free] *CS202: Discrete Structures*, by Saylor Academy. Link: `https://learn.saylor.org/course/view.php?id=67`. **Note**: This course uses only free, open-source materials.
Prerequisites:	(CISC 1110 or CISC 1115 or (both CISC 1113 and CISC 1114) or CISC 1170 or CISC 1180 or CISC 1215) and (MATH 1011 or MATH 1012 or assignment to MATH 1201)
Tools/Resources:	Brightspace; Access to a computer (OS doesn't matter); A pdfLaTeX compiler or Overleaf; Adobe Acrobat Reader DC

1.1 Course Description

`(3 credits) Elementary set theory, functions, relations, and Boolean algebra. Switching circuits, gating networks. Definition and analysis of algorithms.`

`Applications of graph theory to computer science. Related algorithms. Introduction to combinatorial computing and counting arguments. Introduction to error analysis.` (Taken from CUNYFirst.)

1.2 Course Objectives

By the end of this course, you will master the following skills:

- Knowledge and usage of multiple math topics and tools that will benefit you as a Computer Scientist.
- Typesetting math with LaTeX and generation of organized and well-styled PDF documents.
- Simple proofs of mathematical statements (mathematical induction, indirect arguments) and logical propositions (including quantifiers).
- Functional and relational properties (one-to-one, onto, reflexive, symmetric, transitive, equivalence, partial ordering), and operations (composition, transitive closure).
- Counting principles, countable and uncountable sets. Basic probability theory and applications.
- Big-$\mathcal{O}$ Notation. Recursive definitions and solutions of simple recurrence relations.
- Fundamental concepts of set theory and Boolean Algebra.
- Application of matrices, graphs, and trees as data structures/containers.
- Independent searching and verbal expression of answers based on given sources or your opinion.

Please refer to the Required Electronic Tools and Resources section at the end of the full version of the syllabus for information about how to obtain the software required for this course (for free, of course.)

2 Grading Components

The course's grade is influenced by the following components:

Attendance	15%
Participation	30%
Midterm	25%
Final	30%
Extra Credit	5%

3 Grades

Students will receive a letter grade for the course according to the following score distribution established by CUNY:

<60	**60-62**	**63-66**	**67-69**	**70-72**	**73-76**	**77-79**	**80-82**	**83-86**	**87-89**	**90-92**	**93+**
F	D-	D	D+	C-	C	C+	B-	B	B+	A-	A

A grade of A+ will be granted for numerical grades of 97 or higher after all extra credit points you received are applied to the grade.

4 Important Brooklyn College Policies

4.1 Center for Student Disability Services

The Center for Student Disability Services (CSDS) is committed to ensuring students with disabilities enjoy an equal opportunity to participate at Brooklyn College. In order to receive disability-related academic accommodations students must first be registered with CSDS. Students who have a documented disability or suspect they may have a disability are invited to schedule an interview by calling $(718) - 951 - 5538$ or emailing Josephine.Patterson@brooklyn.cuny.edu. If you have already registered with CSDS, email Josephine.Patterson@brooklyn.cuny.edu or testingcsds@brooklyn.cuny.edu to ensure the accommodation email is sent to your professor.

If the CSDS decides that you are eligible for a time extension on exams (such as time-and-a-half or double-time,) you will still take the exam with the rest of the class at the day + time that is listed in the Schedule section below and will be fully accommodated: you will receive the time extension you are eligible for. That is, you won't take the exam at the CSDS test center.

4.2 Nonattendance Because of Religious Beliefs

The Brooklyn College undergraduate Bulletin for the years 2025 – 2026 states:

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The New York State Education Law provides that no student shall be expelled or
refused admission to an institution of higher education because he or she is unable
to attend classes or participate in examinations or study or work requirements
on any particular day or days because of religious beliefs.  Students who are
unable to attend classes on a particular day or days because of religious beliefs
will be excused from any examination or study or work requirements.  Faculty must
make good-faith efforts to provide students absent from class because of religious
beliefs equivalent opportunities to make up the work missed; no additional fees
may be charged for this consideration.
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Based on the description above, if you are incapable of attending a class because of religious observance, you should e-mail the instructor at least 48 hours before that class so that proper accommodations could be made. If this is an exam day, we will schedule a make-up exam when it is convenient to you, and if an assignment is due, the due date will be extended, and the instructor will tell you when the new due date is.

4.3 Brooklyn College Policy on Academic Integrity

The faculty and administration of Brooklyn College support an environment free from cheating and plagiarism. Each student is responsible for being aware of what constitutes cheating and plagiarism and for avoiding both.

The complete text of the CUNY Academic Integrity Policy can be found at this site:

https://www.cuny.edu/about/administration/offices/legal-affairs/policies-resources/academic-integrity-policy/

If a faculty member suspects a violation of academic integrity and, upon investigation, confirms that violation, or if the student admits the violation, the faculty member MUST report the violation. Students should be aware that faculty may use plagiarism detection software.

This means that if you cheat on a test or assignment, the instructor of this course MUST file a report which will initiate academic penalties. Additionally, the assignment in which you cheat will get an unfortunate score of 0.

4.4 Brooklyn College Bereavement Policy

Students who experience the death of a loved one should refer to:

https://www.brooklyn.edu/policies/bereavement/

4.5 Brooklyn College Library

New student? Returning to campus? Looking for materials for your class or research? Check out the plethora of resources that the Brooklyn College Library is providing to you:

https://library.brooklyn.cuny.edu/resources/

You will certainly find something useful there!

4.6 More Information: Bulletin

For more information about the policies of Brooklyn College and other essential information, please refer to the Bulletin, which you can find on the following web-page:

https://www.brooklyn.edu/registrar/bulletins/

5 Important Dates

August 28 (Fr): Start of Fall 2026 Term
August 31 (Mo): First lecture of CISC 2210, section MW3

September 03 (Th): Registrar drops everyone waitlisted for Fall 2026 courses
September 03 (Th): Last day to add a course
September 07 (Mo): Labor Day: College Closed
September 11 – 13 (Fr – Su): No classes scheduled
September 18 (Fr): Grade of W is assigned for officially withdrawing from a course
September 21 (Mo): No classes scheduled
October 12 (Mo): Columbus Day: College Closed
October 13 (Tu): Conversion Day: Classes follow Monday schedule
November 06 (Fr): Last day to withdraw from a course with a grade of W
November 08 (Su): No classes scheduled
November 25 (We): Thanksgiving Eve: No classes scheduled
November 26 – 27 (Th – Fr): Thanksgiving: No classes scheduled
November 28 (Sa): Thanksgiving: No classes scheduled
December 14 (Mo): Last day of classes!
December 15 – 21 (Tu – Mo): Week of Final Examinations for the Fall 2026 Term
December 28 (Mo): Fall 2026 Course Grades will be Posted on CUNYfirst by 11:59 PM.

Please refer to the Brooklyn College Academic Calendar for the Fall 2026 semester to view other important dates not mentioned above:

https://www.brooklyn.edu/registrar/academic-calendars/#fall

6 Schedule

Note that the schedule below is tentative; if changes are made, the instructor will notify you and will post the updated syllabus/schedule on Brightspace.

Week	Date	Topics, Exams, and Assignment Deadlines
1	08/31 (Mo)	**Welcome!** Syllabus Review • Read the Abridged Syllabus before this lecture
	09/02 (We)	Topic 1: Discrete Structures: Overview • Read slides 1 – 16 of Topic 1, inclusively, before this lecture
2	09/07 (Mo)	College Closed: **No CISC 2210, MW3 lecture!**
	09/09 (We)	Topic 1: Discrete Structures: Overview – Cont' • Read slides 17 – 28 of Topic 1, inclusively, before this lecture
3	09/14 (Mo)	Topic 2: Sets & Sequences • Read slides 1 – 20 of Topic 2, inclusively, before this lecture
	09/16 (We)	Topic 2: Sets & Sequences – Cont' • Read slides 21 – 54 of Topic 2, inclusively, before this lecture
4	09/21 (Mo)	No classes scheduled: **No CISC 2210, MW3 lecture!**
	09/23 (We)	Topic 2: Sets & Sequences – Cont' • Read slides 55 – 66 of Topic 2, inclusively, before this lecture

Week	Date	Topics, Exams, and Assignment Deadlines
5	09/28 (Mo)	Topic 3: Logic & Simple Proof Methods • Read slides 1 – 18 of Topic 3, inclusively, before this lecture
	09/30 (We)	Topic 3: Logic & Simple Proof Methods – Cont' • Read slides 19 – 30 of Topic 3, inclusively, before this lecture
6	10/05 (Mo)	Topic 3: Logic & Simple Proof Methods – Cont' • Read slides 31 – 48 of Topic 3, inclusively, before this lecture
	10/07 (We)	Topic 3: Logic & Simple Proof Methods – Cont' • Read slides 49 – 67 of Topic 3, inclusively, before this lecture
7	10/12 (Mo)	College Closed: **No CISC 2210, MW3 lecture!**
	10/13 (Tu)	Conversion Day: **We have a lecture today!** Topic 4: Relations • Read slides 1 – 16 of Topic 4, inclusively, before this lecture
8	10/14 (We)	Topic 4: Relations – Cont' • Read slides 17 – 28 of Topic 4, inclusively, before this lecture
9	10/19 (Mo)	Topic 5: Functions • Read slides 1 – 17 of Topic 5, inclusively, before this lecture
	10/21 (We)	Topic 5: Functions – Cont' • Read slides 18 – 29 of Topic 5, inclusively, before this lecture
10	10/26 (Mo)	Topic 6: Counting & Combinatorics • Read slides 1 – 19 of Topic 6, inclusively, before this lecture
	10/28 (We)	Topic 7: Discrete Probability • Read slides 1 – 11 of Topic 7, inclusively, before this lecture
11	11/02 (Mo)	Topic 7: Discrete Probability – Cont' • Read slides 12 – 20 of Topic 7, inclusively, before this lecture
	11/04 (We)	**Midterm Exam: 03:40 PM – 04:55 PM, at room TBA**
12	11/09 (Mo)	Topic 8: Recurrence & Recursion • Read slides 1 – 9 of Topic 8, inclusively, before this lecture
	11/11 (We)	Topic 8: Recurrence & Recursion – Cont' • Read slides 10 – 20 of Topic 8, inclusively, before this lecture
13	11/16 (Mo)	Topic 8: Recurrence & Recursion – Cont' • Read slides 21 – 31 of Topic 8, inclusively, before this lecture
	11/18 (We)	Topic 9: Boolean Algebra • Read slides 1 – 33 of Topic 9, inclusively, before this lecture
14	11/23 (Mo)	Topic 10: Matrices • Read slides 1 – 17 of Topic 10, inclusively, before this lecture
	11/25 (We)	No classes scheduled: **No CISC 2210, MW3 lecture!**

Week	Date	Topics, Exams, and Assignment Deadlines
15	11/30 (Mo)	Topic 10: Matrices – Cont' • Read slides 18 – 43 of Topic 10, inclusively, before this lecture
	12/02 (We)	Topic 10: Matrices – Cont' • Read slides 44 – 67 of Topic 10, inclusively, before this lecture
16	12/07 (Mo)	Topic 11: Graphs & Trees • Read slides 1 – 26 of Topic 11, inclusively, before this lecture
	12/09 (We)	Topic 11: Graphs & Trees – Cont' • Read slides 27 – 38 of Topic 11, inclusively, before this lecture
17	12/14 (Mo)	Topic 11: Graphs & Trees – Cont' • Read slides 39 – 56 of Topic 11, inclusively, before this lecture
Finals	12/16 (We)	**Final Exam: 03:30 PM – 05:30 PM, at room TBA**

– End of CISC 2210 Abridged Syllabus –